Supervised Model Evaluation

Model Evaluation : model evaluation is the process of using specific metrics and technique to measure how well a machine learning model performs.

Suppose two model make prediction:

Model A = 90% Accuracy
Model B = 75% Accuracy

Q) Can we always say model A is better?

Ans: Ofcourse Not, No! because the correct metric depends on the problem.

∴ Model → prediction → Evaluation → Better Model

Goal : to measure how well model perform on unseen data.

1) Supervised Model evaluation has 2 types:

i) classification Model Evaluation: It predicts classes such as Yes/No, Spam/Not Spam, Disease/No disease etc.

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→ Classification model evaluation using confusion Matrix:

From the table below:

	Actual positive	Actual Negative
predicted positive	TP = 40	FP = 10
predicted Negative	FN = 20	TN = 30

where, TP (True positive) → correctly predicted +ve
TN (True Negative) → correctly predicted -ve
FP (False positive) → Negative predicted as +ve
FN (False Negative) → positive predicted as -ve

~~Exp~~ Important Classification Matrix

1) Accuracy = overall percentage of correct prediction

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

2) Precision = of all predicted positives, how many are actually positive?

$$\text{precision} = \frac{TP}{TP + FP}$$

3) Recall = of all actual positive, how many did the model correctly identify

$$Recall = \frac{TP}{TP + FN}$$

4) F1-Score = Balance between precision and recall

$$F_1 = 2 \times \frac{Precision \times Recall}{Precision \times Recall}$$

Eg:

If a disease prediction model gives,

TP = 80

TN = 90

FP = 10

FN = 20

Then,

Accuracy = 85%

Precision = 88.9%

Recall = 80%

F1-Score ≈ 84.2%

ii) Regression Model Evaluation: It predicts continuous ~~value~~ numerical value such as house price, sales, ~~b~~ temperature, salary etc

→ This model predicted value is compared with actual value using ~~et~~ the error.

$$\text{Error} = \text{Actual} - \text{predicted}$$

Important regression Metrics

1) MAE (Mean Absolute Error): Average absolute difference betwn actual and predicted value.

$$MAE = \frac{1}{n} \sum |y - \hat{y}|$$

lower MAE is better model

2) MSE (Mean Square Error): Average of squared errors.

$$MSE = \frac{1}{n} \sum (y - \hat{y})^2$$

lower MSE is better model

3) RMSE (Root Mean Squared Error) :
square root of MSE.

$$RMSE = \sqrt{MSE}$$

It is usefull because it's unit is the same as the target variable.

4) R^2 Score (coeficient of Determination) :
Shows how much variation in the target variable is explained by the model.

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

$R^2 = 1 \rightarrow$ perfect model
$R^2 = 0 \rightarrow$ Model explain no improvement over the mean baseline
Higher $R^2 \rightarrow$ generally better